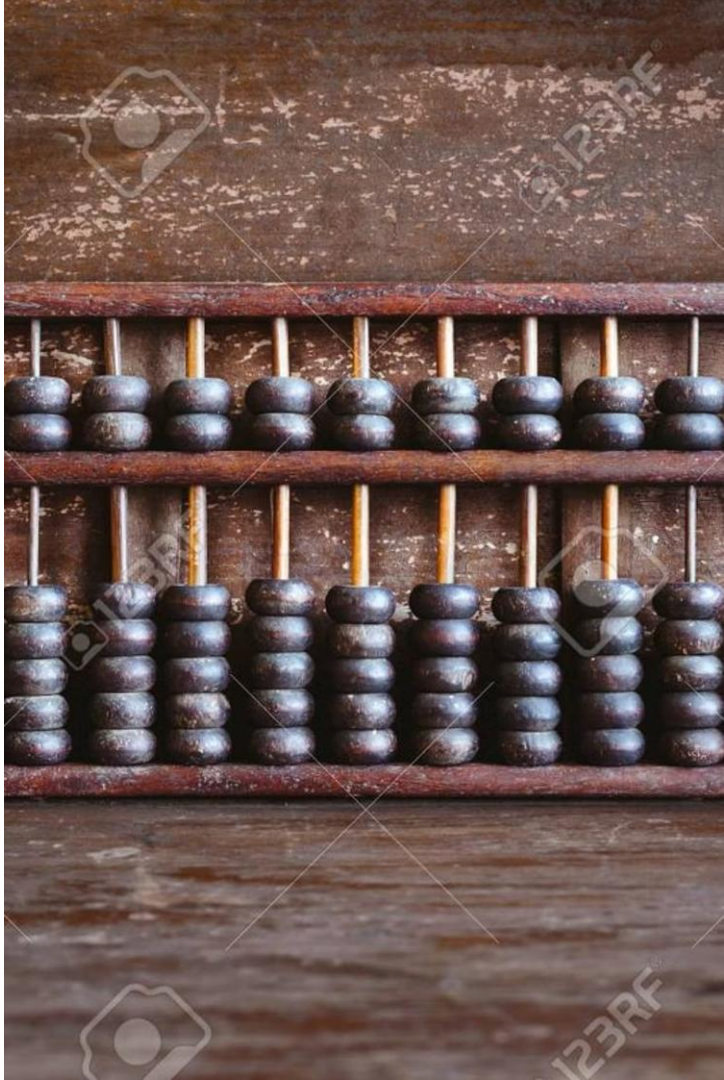
From Abacus to AI

A Journey Through Computing History — tracing humanity's relentless drive to calculate, automate, and innovate across five thousand years.



The Dawn of Calculation

*Long before silicon chips, humans counted with beads. The **abacus** — first developed in ancient Mesopotamia around 2700 BCE — became the world's first calculating tool, spreading through China, Greece, and Rome. Its elegant row-and-bead design allowed merchants and scholars to perform addition, subtraction, and even multiplication with remarkable speed.*

~2700 BCE

Sumerian abacus emerges in Mesopotamia

500 BCE

Greek and Chinese variants developed independently

Still in Use

Suanpan & soroban used in Asia today



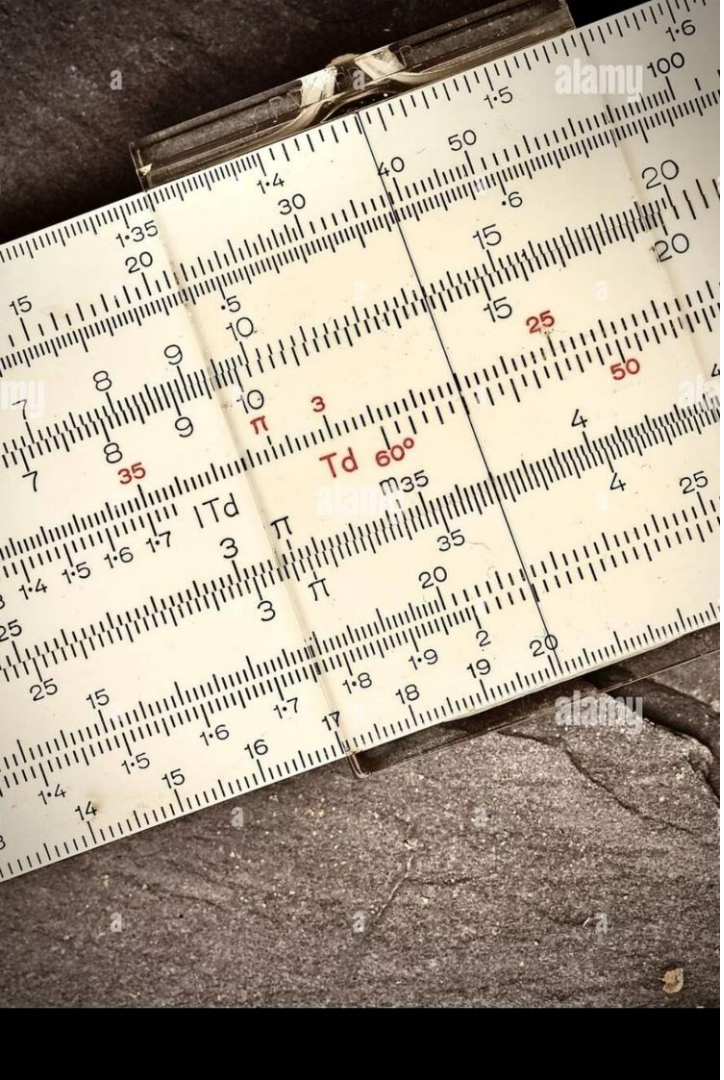


1617

Napier's Bones & Early Logarithms

*Scottish mathematician **John Napier** revolutionized multiplication with two landmark inventions. His calculating rods — "Napier's Bones" — used carved ivory or wood strips to break complex multiplications into simple additions. Simultaneously, his invention of **logarithms** transformed astronomy and navigation by reducing laborious calculations to table lookups.*

Napier's logarithm tables remained the computational backbone of science for over 350 years.



The Power of Analogue: The Slide Rule

*Born from Napier's logarithms, the **slide rule** became the engineer's most trusted companion for over 300 years. By sliding logarithmic scales against each other, users could multiply, divide, and compute trigonometric functions in seconds — entirely without electricity.*

Invented ~1622

William Oughtred combined two Gunter scales, creating the first true slide rule

Apollo Engineers Used It

NASA engineers relied on slide rules to calculate trajectories for the Moon landings

Retired in 1972

The pocket calculator finally made the slide rule obsolete after 350 years of service

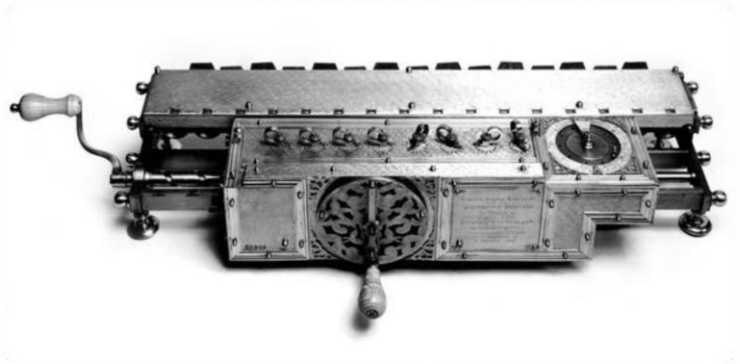
MECHANICAL COMPUTATION

Towards Automation

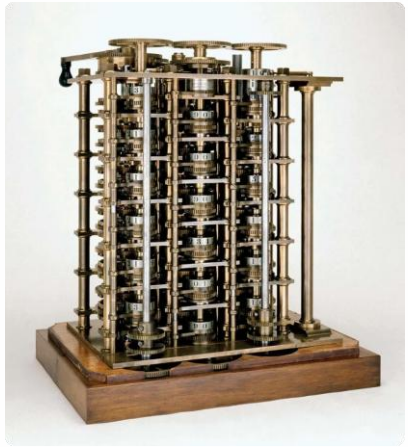
Blaise Pascal *built the Pascaline in 1642 — the first mechanical calculator, designed to help his tax-collector father. Gears and wheels carried digits automatically between columns. Thirty years later,* ***Gottfried Wilhelm Leibniz*** *improved the concept with his Stepped Reckoner in 1694, capable of multiplication and division through a novel stepped-drum mechanism.*

> *Both machines were handcrafted to extraordinary precision — a true feat for 17th-century craftsmen.*

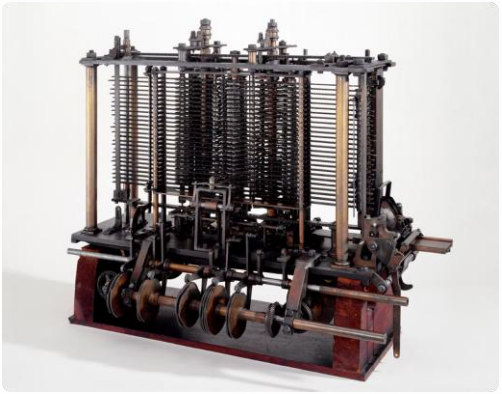




Diffference Engine



Analytical Engine



The Visionaries: Babbage & Lovelace

Charles Babbage

*In 1837, Babbage designed the **Difference Engine** in 1822 AD and **Analytical Engine** — a mechanical general-purpose computer with an arithmetic logic unit, conditional branching, loops, and integrated memory in 1833 AD. Though never fully built in his lifetime, its architecture anticipated the modern computer by over a century.*

Ada Lovelace

***Ada Lovelace** translated and annotated Babbage's work, adding her own notes three times longer than the original text. Her algorithm for computing Bernoulli numbers is recognized as the **world's first computer program**, earning her the title of the first programmer.*



1890S – 1940S

The Electromechanical Era

Herman Hollerith's *punch-card tabulator transformed the 1890 US Census, processing data in weeks rather than years. His company eventually became* ***IBM****. The 1930s–40s brought electromechanical relay computers — Konrad Zuse's Z3 (1941) was the first programmable, fully automatic digital computer, using binary arithmetic and floating-point logic.*

→ Hollerith Tabulator (1890)

Punch-card data processing at industrial scale

→ Zuse Z3 (1941)

First programmable binary computer using relays

The Rise of Electronics

The electronic age began with a room-sized marvel.

ENIAC (1945)

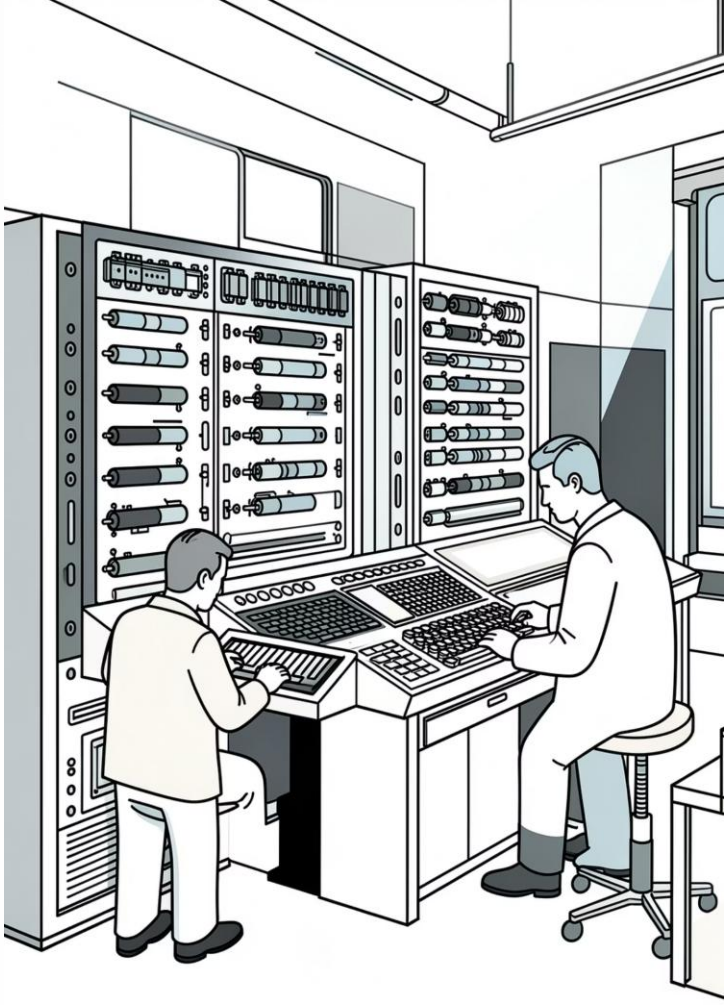
30 tons, 18,000 vacuum tubes — the first general-purpose electronic computer. Calculated artillery trajectories 1,000× faster than humans.

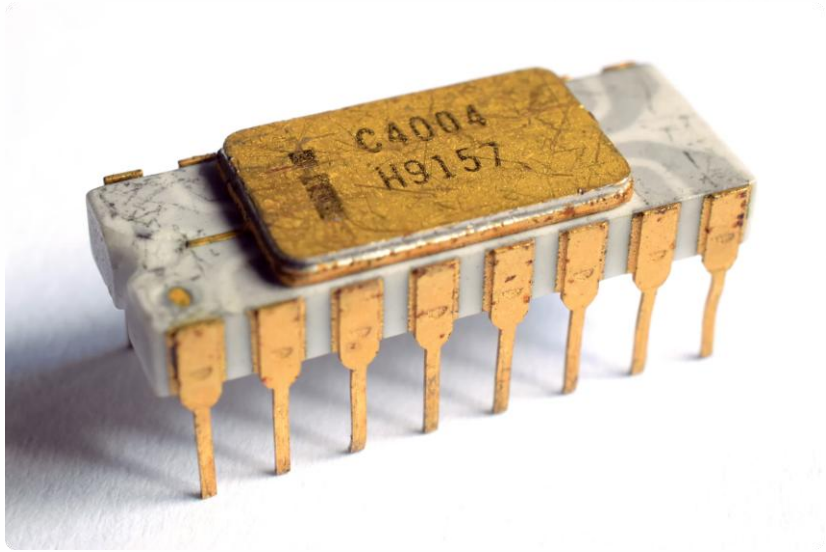
UNIVAC (1951)

First commercial computer sold to the US Census Bureau. Famously predicted Eisenhower's 1952 election victory on live TV.

Transistor (1947)

Invented at Bell Labs, the transistor replaced vacuum tubes — smaller, faster, cooler. It ignited the microelectronics revolution that defines computing today.





1971 – PRESENT

The Modern Age

*The **Intel 4004** (1971) — the first microprocessor — crammed an entire CPU onto a single chip. This sparked the personal computer revolution. By 1991, Tim Berners-Lee's **World Wide Web** connected humanity in an information network no one could have imagined. Smartphones, cloud computing, and social media followed, placing supercomputers in every pocket.*

The Future of Computing

The next frontier is already unfolding — faster, smarter, and stranger than anything before.



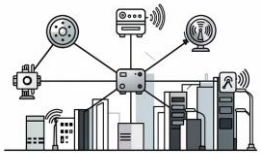
Artificial Intelligence

Large language models and deep learning are redefining what machines can do — from creativity to scientific discovery.



Quantum Computing

Leveraging quantum superposition, tomorrow's computers will solve problems that would take classical machines millions of years.



Edge & Neuromorphic

Computing moves to the edge of networks and mimics the human brain — enabling real-time intelligence everywhere.